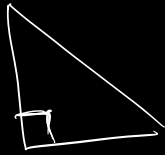
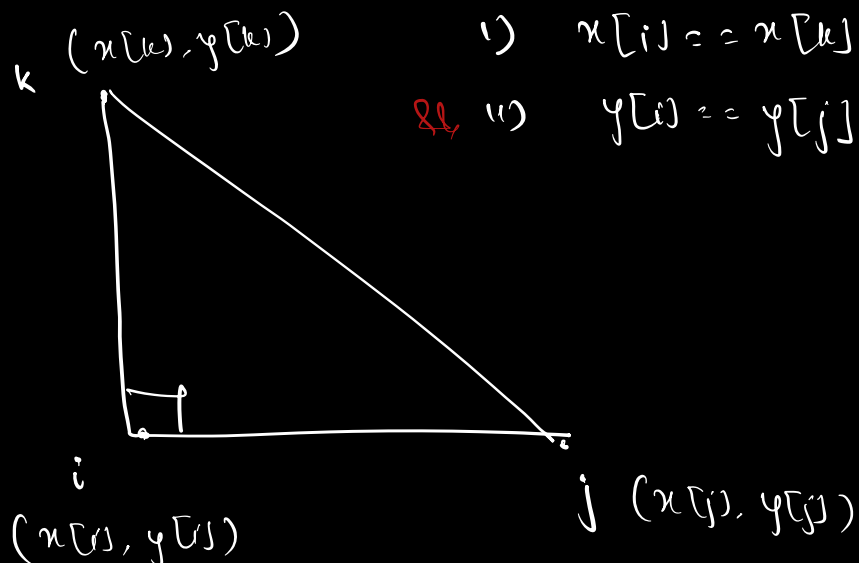
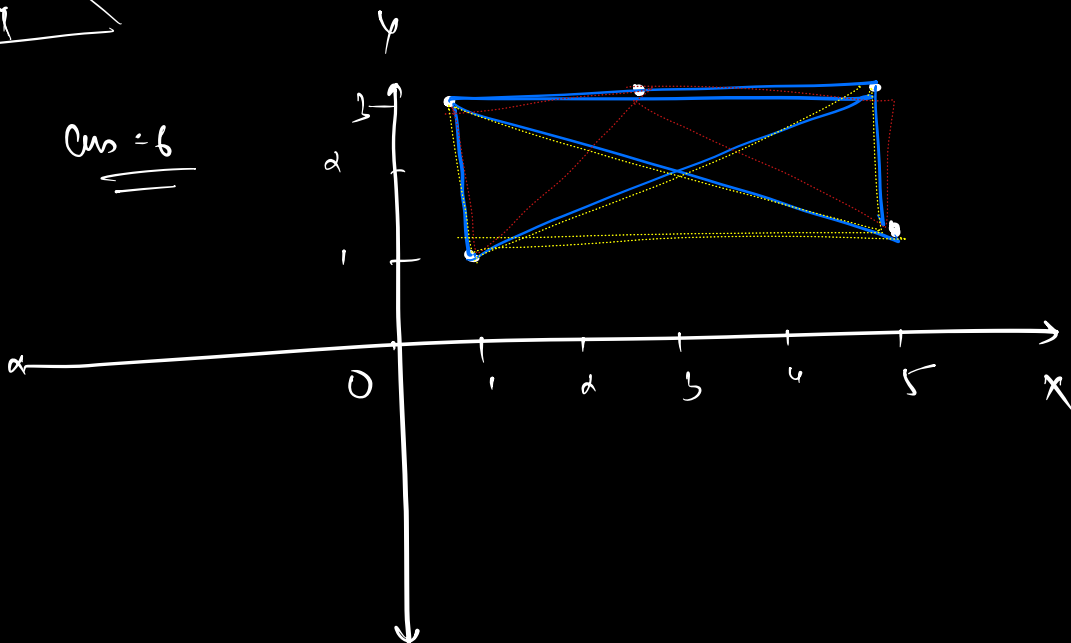


Q1. Given coord of N pts. on a 2D plane. Count no. of right angle triangle that can be formed using the pts. such that the sides of right angle triangles are ||r to x and y axis.

	0	1	2	3	4
$x \Rightarrow$	1	5	3	1	5
$y \Rightarrow$	1	1	3	3	3



Ans = 6



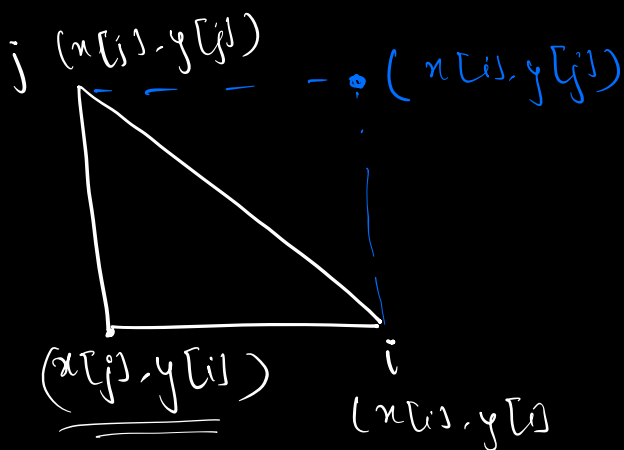
```

for( i=0; i<N; i++) {
    for( j=0; j<N; j++) {
        if( i==j)
            continue
        for( k=0; k<N; k++) {
            if( k==i || k==j)
                continue
            if( x[i]==x[k] && y[i]==y[j]) {
                cout << p << endl;
            }
        }
    }
}

```

TC $\Rightarrow O(N^3)$

SC $\Rightarrow O(1)$



```

for( i=0; i<N; i++) {
    for( j=0; j<N; j++) {
        if( check(x[j], y[i]))
            continue;
        if( check(x[i], y[j]))
            continue;
    }
}

```

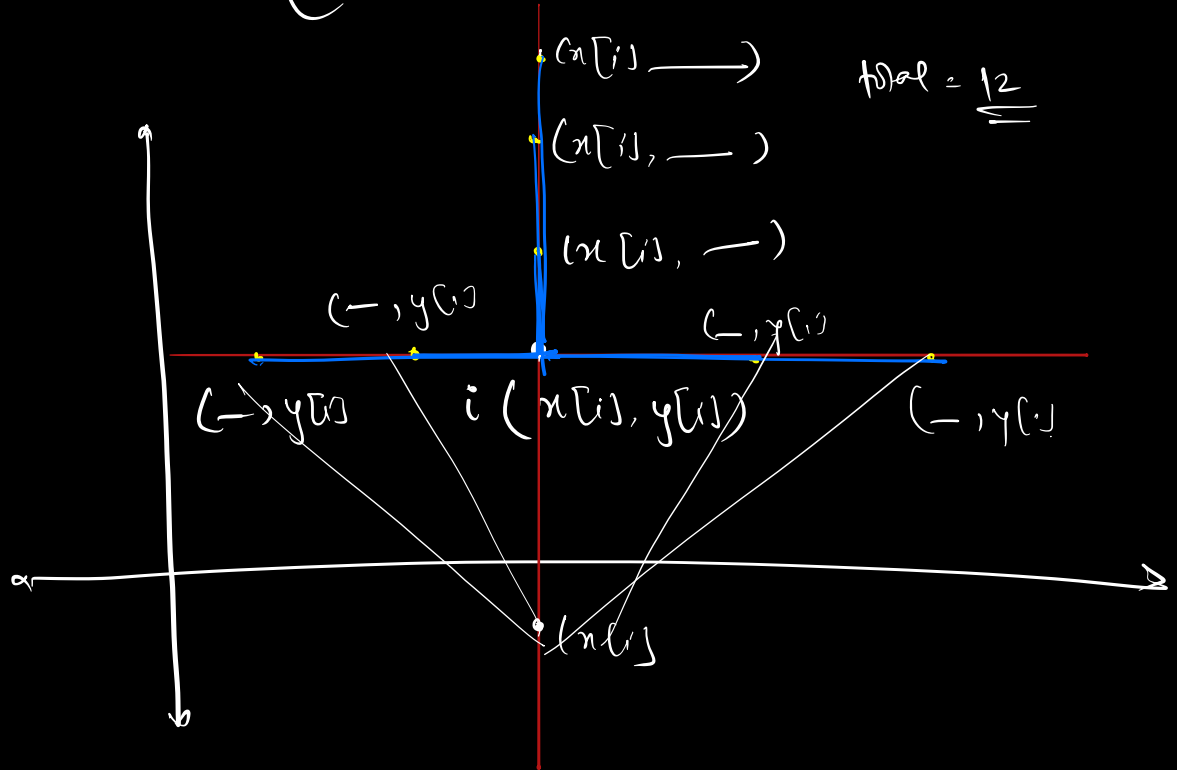
}

	0	1	2	3	4	5
$x \Rightarrow$	1	5	3	1	5	2
$y \Rightarrow$	1	1	3	3	3	4

$TC \Rightarrow O(N^2)$
 $SC \Rightarrow O(N)$

$Set \Rightarrow$

1-1	1-3
5-1	5-3
3-3	2-4



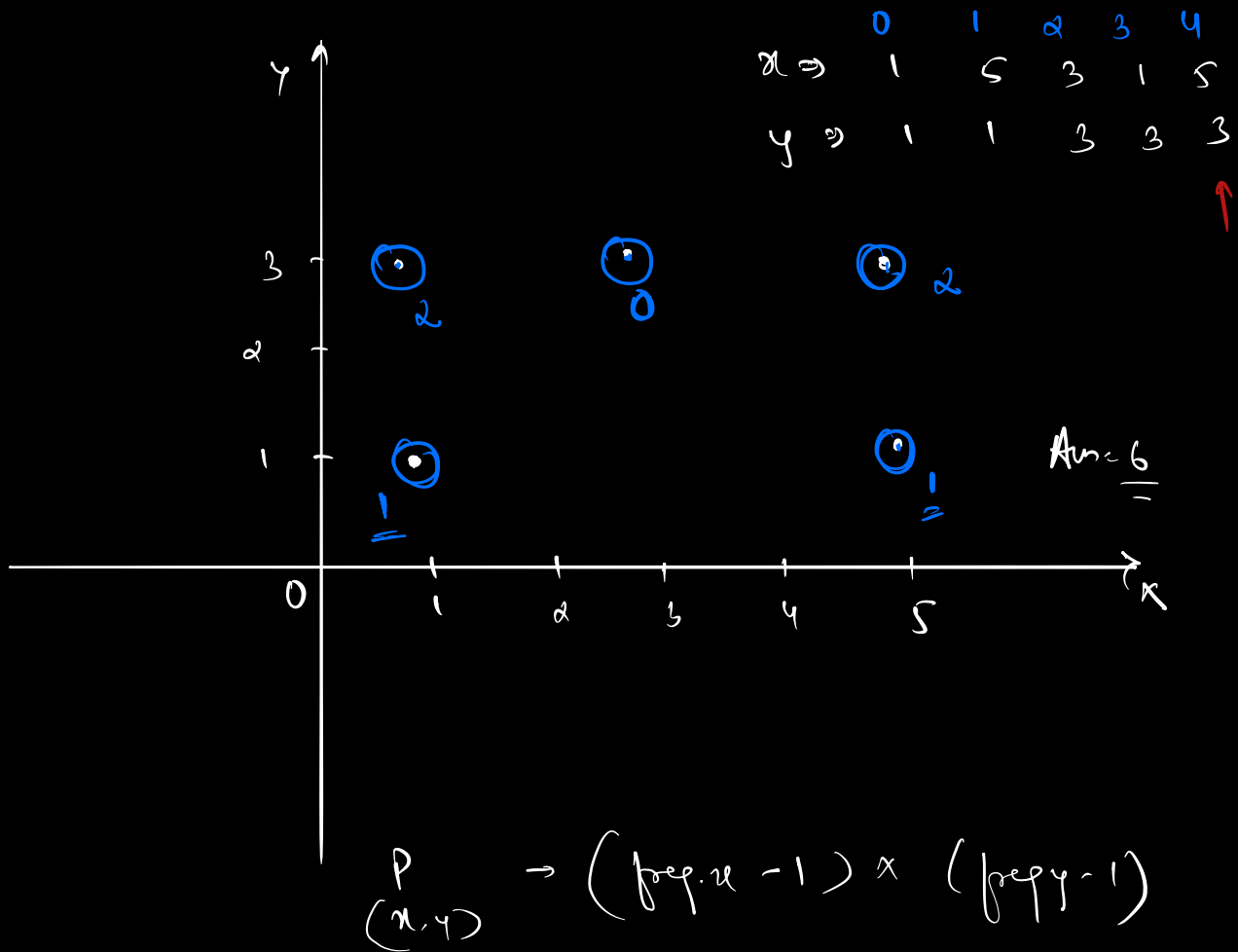
$x \Rightarrow$

1 $\rightarrow$ 2
5 $\rightarrow$ 2
3 $\rightarrow$ 1

 $y \Rightarrow$

1 $\rightarrow$ 2
3 $\rightarrow$ 3

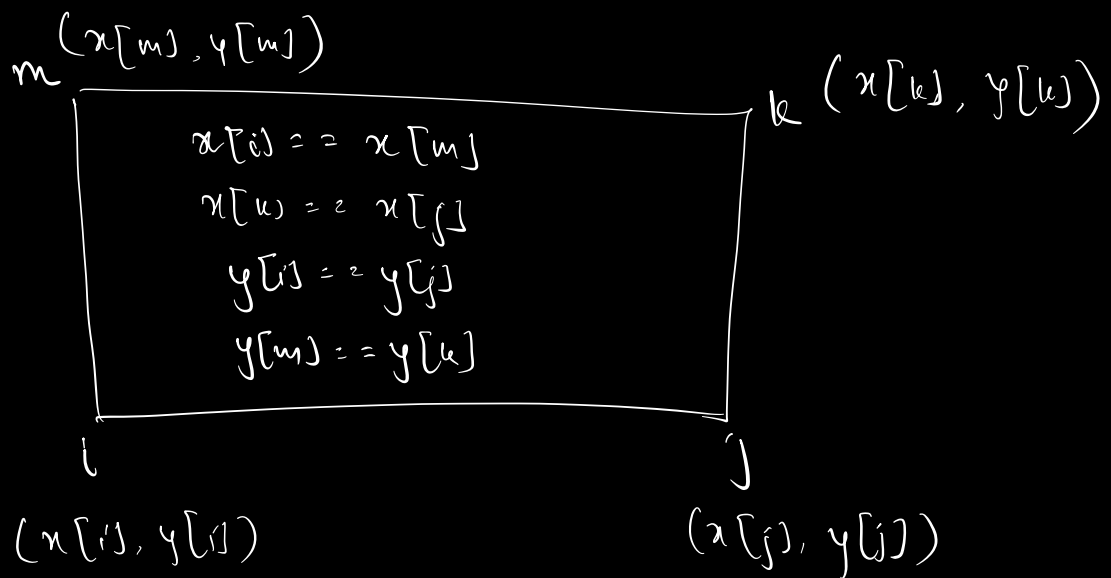
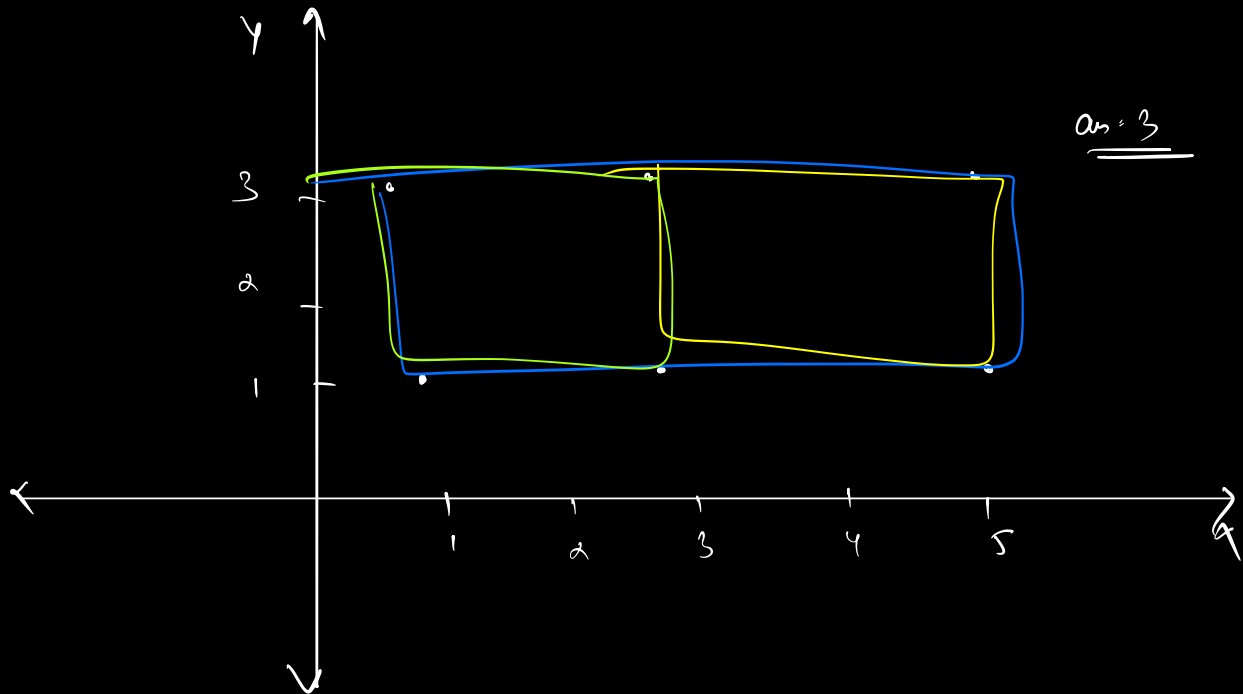
$\left\{ \begin{array}{ll} 1 \times 1 = 1 & 1 \times 2 = 2 \\ 1 \times 1 = 1 & 1 \times 2 = 2 \\ 0 \times 2 = 0 & \end{array} \right.$



TC $\Rightarrow$ $O(N^2)$
 SC $\Rightarrow$ $O(N)$

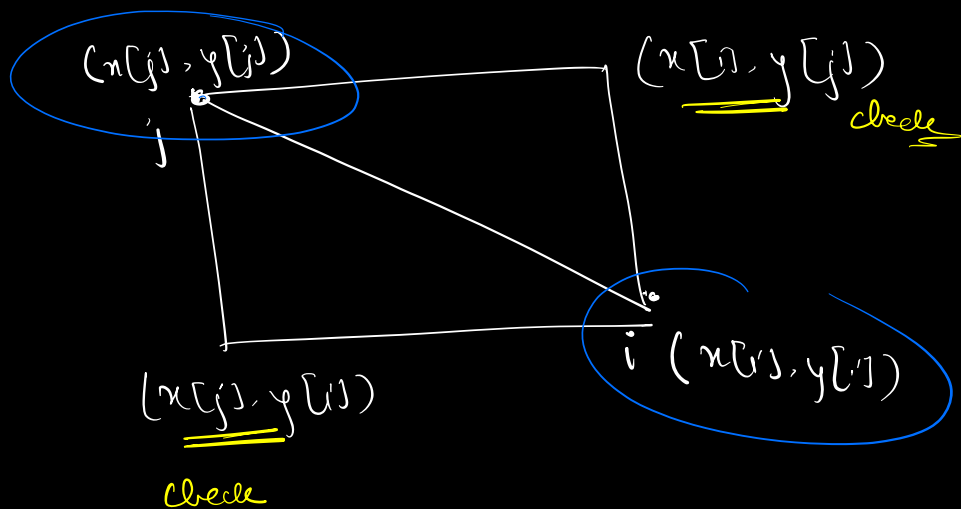
Q1. Given coord of N pts. on a 2D plane. Count no. of rectangle that can be formed using the pts such that the sides of rectangle are ||r to x and y axis.

$x \Rightarrow$ 1 3 5 1 3 5
 $y \Rightarrow$ 1 1 1 3 3 3



$$\underline{\underline{O(N^4)}} \rightarrow \underline{\underline{O(N^3)}}$$

4 pm. 3 pm.



$O(N^2)$ [2 pths.]

~~$O(N)$~~

text $\leftarrow S \Rightarrow$ a b **c b a** b n a $\rightarrow N$
pattern $\leftarrow t \Rightarrow$ c b a $\rightarrow k$

$O(N * k)$ $\rightarrow$ check all substrings



a	b	c	d	e	f	_____
↓	↓	↓	↓	↓	↓	
0	1	2	3	4	5	

a	b	c	d	e	f	hash	
						→	15
						sum of	
						values of	
						alphabet	

b	c	a	d	e	f	hash	
						→	15

d	f	e	c	a	b	hash	
						→	15

e	f	d	a	b	c	hash	
						→	15

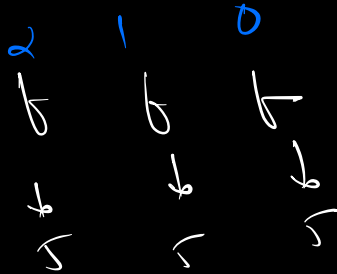
f	f	f	→	15
---	---	---	---	----

5	4	3	2	1	0
4	1	3	6	9	2

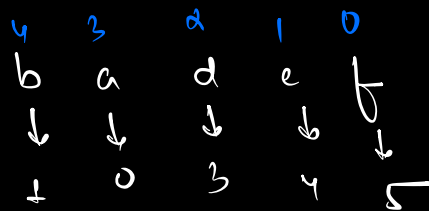
$$\Rightarrow 4 \times 10^5 + 1 \times 10^4 + 3 \times 10^3 + 6 \times 10^2 + 9 \times 10^1 + 2 \times 10^0$$

5	4	3	2	1	0
a	b	c	d	e	f
↓	↓	↓	↓	↓	
0	1	2	3	4	

$$\Rightarrow 0 \times 26^4 + 1 \times 26^3 + 2 \times 26^2 + 3 \times 26^1 + 4 \times 26^0$$



$$\Rightarrow 5 \times 26^2 + 5 \times 26^1 + 5 \times 26^0$$



$\Rightarrow i$ 0 1 2 3 4

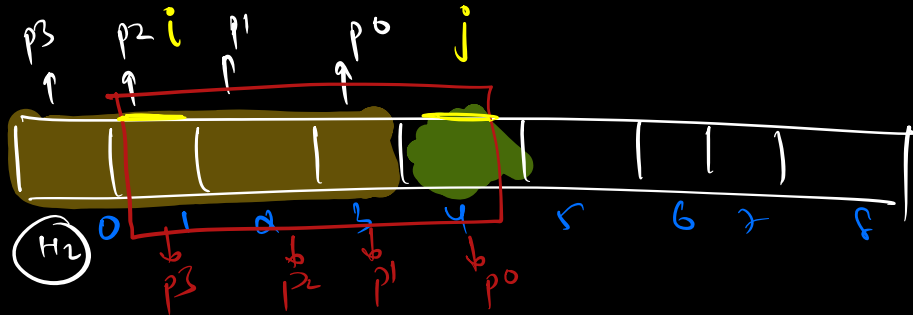
$$1 \times 26^4 + 0 \times 26^3 + 3 \times 26^2 + 4 \times 26^1 + 5 \times 26^0$$

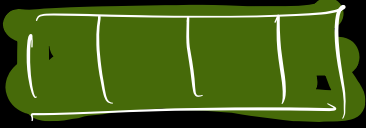
$$\left(\sum \text{value}[s[i]] \times p^{N-i-1} \right) \% M$$

$$p \geq 26$$

$$M = \underline{\underline{10^9 + 7}}$$

$$\text{Hash func} \rightarrow \left(\sum \text{val}[s[i]] \times p^{N-i-1} \right) \% M$$



pattern  $\rightarrow$ H_1

$$\text{yellow} = s[0] \times p^3 + s[1] \times p^2 + s[2] \times p^1 + s[3] \times p^0$$

$$H_2 = (\text{yellow} - s[0] \times p^3) \times p + s[4]$$

$$\text{red} = s[1] \times p^3 + s[2] \times p^2 + s[3] \times p^1 + s[4] \times p^0$$

$$\text{yellow} \Rightarrow (s[1] \times p^2 + s[2] \times p^1 + s[3] \times p^0) \times p + s[4]$$

$$H = \left[H - s[i-1] \times p^{k-1} \right] \times p + s[j]$$

$$\left(\frac{10^5}{10^9} \times 100 \right) = \underline{\underline{0.01\%}}$$

$$\left(\frac{10^5}{10^{12}} \times 100 \right) = \underline{\underline{0.0001\%}}$$

Robin Karp Algo
